WACV #0000

Viewpoint Invariance Is a Theorem, Not a Fit: An $SE(3)$-Equivariant Recurrence for Rehabilitation Exercise Assessment

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Abstract

A clinical screening instrument must return the same patient the same score from a different room. We show that for skeleton-based rehabilitation assessment this per-prediction guarantee is available by construction: an $SE(3)$-equivariant recurrent model whose read-out is invariant to camera viewpoint as a theorem*, certified to the fp64 roundoff floor ($\sim 10^{-16}$) over Haar-random $SO(3)$ — of which any camera relocation, including the $45°$ NTU RGB+D Cross-View displacement, is one instance. On clean frontal data no model separates from another — ours, a point-cloud-transformer baseline, and a hand-crafted-invariant GRU tie within a 0.33 MAD nondeterminism floor — so the benchmark's aggregate metric is saturated and the decisive axis lies elsewhere: the augmented baseline matches us in aggregate accuracy while still swinging an individual patient's score by up to 20.1 MAD (of 50) under camera rotation, against our 9×10^{-6}. We then map the accuracy–exactness–robustness–cost frontier and find that* no model on it dominates*: the point-cloud baseline is both the most accurate and the most node-robust, a hand-crafted-invariant control is the cheapest and exactly invariant yet collapses under sensor-node failure, and a lighter $E(n)$ encoder behind the identical invariant cut is measurably more brittle. Ours is the only point that is exact* by theorem *while clearing every stress up to three simultaneous node failures, at $7.4\times$ fewer parameters. We replicate the viewpoint and node-failure properties on a second corpus (REHAB24-6), and measure what the theorem costs on the edge: an int8 activation-quantization cliff of only 0.05 MAD — 0.005 of a between-subject SD on this corpus, and one to two orders of magnitude below MCID thresholds on comparable rehabilitation scales (Sec. 4) — and a causal streaming variant at 112 fps (GPU, batch 1). We report our negative results — a null irregular-sampling regime and an unrewarded parity restoration — with equal prominence.*

1. Introduction

Automated assessment of rehabilitation exercises from skeletons promises objective remote monitoring [1]. The dominant regime, however, trains and tests on clean frontal recordings from one sensor and ranks models by a single aggregate error against a clinician's score [2–5]. It rewards capacity and penalizes nothing a deployed device meets: a camera against a different wall, a tracking node that freezes, a dropped frame.

This paper argues that the *regime*, not the architectures, is the problem. On clean frontal KIMORE no architecture separates from another: our model, the point-cloud transformer, and a GRU on six hand-crafted invariant features are statistically indistinguishable — the benchmark's aggregate metric is saturated. We can say so precisely because we first measured what a difference must exceed to be one: training here is not bit-reproducible (nondeterministic cuDNN and scatter-add kernels), and repeated fits of an *identical* configuration differ by ≈ 0.33 MAD — larger than every clean-accuracy gap between the leading models. A single-run comparison on this benchmark reports its own seed. Worse, auditing the reference transformer on the commonly used single-exercise slice, optimistic epoch selection is worth $+1.2 \pm 0.4$ MAD (18.6% over three seeds), and its honest number carries no reliable subject-level signal: the paired subject-cluster bootstrap gives $\Delta = -0.44$ (per-seed $[-1.07, +0.20]$), clearing zero in only one of three seeds — the slice's limit, not the model's. We therefore pool the five exercises, keep splits subject-disjoint, and gate every claim on clearing a per-exercise mean-predictor floor — a model that ignores its input has zero degradation and would win every robustness plot. Pooled, every model we test clears it (Tab. 3).

With clean accuracy neutralized, the question becomes what a structural guarantee buys. We formulate assessment as an *invariant read-out of an equivariant representation*: a steerable message-passing encoder produces per-joint features in $O(3)$ irreducible representations, an explicit projection onto a generating set of invariants discharges the symmetry, and a recurrence consumes the invariant code with the sensor's actual inter-arrival times (a capability KIMORE does not reward, Sec. 4; kept because it is free and demonstrably real off this corpus). Invariance of the score under any rigid camera motion is then a *theorem* (Proposition 1), true for arbitrary weights, not a behaviour induced by augmentation.

We contribute three things, none of them new invariance theory (scalar contractions of Wigner-D-covariant features are invariant by construction [10–12]). An *end-to-end certification* protocol: eight numerical gates that survive the real steerable layers, the adaptive solver and int8 quantization. An *evaluation methodology* the field lacks: a mean-predictor floor below which no gap is interpretable, and per-sequence degradation reported as a distribution rather than a mean. And a *deployment* analysis: a precision budget and a causal streaming read-out. The distinction bites: a rotation-augmented baseline has a flat *aggregate* accuracy curve across azimuth — by the field's metric it has solved viewpoint — yet its *per-sequence* degradation averages 3.0 MAD and reaches 20.1 in the tail — individual scores swinging up to two fifths of the fifty-point scale as the camera moves, the errors merely cancelling in the mean. Ours move by 9×10^{-6}; we report degradation and accuracy side by side throughout.

The tie with the hand-crafted model leaves an obligation elegance cannot discharge: if a plain GRU on named features is exactly as accurate *and* as viewpoint-invariant, the steerable machinery is unjustified. We discharge it against *that* model specifically. When Kinect nodes freeze — the characteristic consumer-depth failure — the hand-crafted model fails outright: a *single* dead joint drives it from 6.31 to 9.78 MAD, through the floor, because its features *name* the joints they depend on; the graph encoder pools learned messages and dilutes the failure across 25 joints. That is one pairwise comparison, not an axis we own: the point-cloud baseline is more node-robust than either of us, and our case rests on *where* we sit on the accuracy–exactness–robustness–cost frontier (Tab. 6). We also treat the guarantee as a *systems* property: it survives to int8 (only activation-, not weight-rounding perturbs it) and converts to strictly causal streaming at 8.9 ms/frame. Finally we report the two places the structural story does not pay — native consumption of irregular arrivals, which a fixed grid beats at every drop level here, and a restored parity channel, principled but unrewarded on healthy-form exercises.

Contributions. (i) A protocol audit that establishes what the benchmark can measure — an 18.6% epoch-selection inflation on the reference model, a 0.33 MAD nondeterminism floor below which no accuracy gap is interpretable, and a gated pooled protocol under which every model clears the floor. (ii) Exact viewpoint invariance, *certified* by eight numerical gates (E1–E8) over Haar-random $SO(3)$ rather than a single azimuth sweep, against a baseline that crosses the floor by 45°. (iii) A Pareto characterisation of the accuracy–exactness–robustness–cost frontier on which no model dominates: the baseline wins clean accuracy *and* node robustness, while ours is alone in being exact by theorem at $7.4\times$ fewer parameters — including, behind the identical invariant cut, evidence that lighter $E(n)$ equivariance is more node-fail brittle. (iv) A second-corpus replication (REHAB24-6) of the viewpoint and node-failure properties. (v) A co-designed deployment path: an int8 precision-budget audit and a causal streaming read-out with an honest time-to-first-score. We also report two negative results, one of them derived.

2. Related Work

Rehabilitation assessment and equivariant models. KIMORE [1] is the standard benchmark; reported results span ST-GCN variants with contrastive objectives [3], cross-modal augmentation [4], dual-stream graph networks [5], and point-cloud transformers [2], with Rehab-Pile [6] (aggregating KIMORE, UI-PRMD [7] and IRDS) the closest antecedent of our protocol discipline. Evaluation is uniformly in-distribution (same sensor, same placement, no simulated failure), and to our knowledge no prior work reports a mean-predictor floor, without which a degradation curve cannot be interpreted.

Published numbers do not enter Tab. 3, because no protocol matches ours. Kuang et al. [5] report MAD 0.10–0.16 in the *same* 0–50 clinical-score units as our subject-disjoint 6.3–6.7, lower by a factor of 39–67; their split is an explicit 8:1:1 *sample-level* stratified one. We recover those units from the published tables (supplement) because the reference implementation standardises its targets and reading it wrong would move the comparison by $8.5\times$. Karlov et al. [3] report Spearman ρ rather than MAD, so we report ρ too: our pooled $\rho \approx 0.6$ sits below the 0.74–0.965 band the literature spans, on a protocol whose mean-predictor null we also state. Abedi et al. [4] report only a relative MAD reduction with no recoverable absolute baseline. Lacking their data and a subject-disjoint rerun, we treat these as a different protocol, not an accuracy ceiling.

Group-equivariant networks impose symmetry as a constraint on the hypothesis class [9]; the steerable family — tensor-field networks [10], $SE(3)$-Transformers [11], `e3nn` [12] — guarantees equivariance for arbitrary weights, while lighter constructions reach $E(n)$-equivariance without spherical harmonics [13] or via vector neurons [14]. We build on [12], but the claim is *not* about steerable machinery: a GRU on hand-crafted invariants attains the same accuracy and viewpoint guarantee at a third of the parameters, and we isolate the one axis on which they differ (Tab. 4).

Invariance by canonicalization, augmentation, and

corruption robustness. A non-equivariant backbone can be made invariant by canonicalization or frame averaging [15], exact only in so far as the frame estimate is — a point estimate whose *conditioning* degrades near covariance degeneracy, which we measure directly (Sec. 4); our cut estimates no frame, being an explicit projection exact for arbitrary weights (Proposition 1). Training-time augmentation flattens *aggregate* error but makes per-prediction errors cancel, not vanish — expressible only if degradation is reported per sequence. For irregular sampling, the standard alternatives are resampling or continuous-time models [18–21]; we implement the Neural CDE variant (adaptive Dormand–Prince solver [22]) as a certified *control* that fails the floor. Occlusion and tracking failure are usually met with masking or attention [8]; we adopt the failure literally, as the discriminating experiment between two otherwise-indistinguishable models.

3. Method

A steerable encoder on the skeleton graph carries per-joint latents in $O(3)$ irreps (n_0 type-0 scalars s_j, n_1 type-1 vectors v_j); feeding *root-relative* coordinates makes translation invariance automatic. The encoder is *equivariant*; the score must be *invariant*.

The invariant cut. The transition is an explicit projection Π onto a generating set of invariants. The parity-even generators (scalars, vector norms, pairwise cosines) are complete for the full $O(3)$ — each is fixed by a reflection — so a cut built from them is blind to *trajectory handedness* (a clockwise and anticlockwise limb circle share every bone length, distance and speed). We complete it with parity-odd pseudo-scalar triple products over unit vectors:

$$\pi_j^{\text{even}} = \Big[\tanh s_j \,\|\, \log(1{+}\|v_j^c\|)_c \,\|\, \langle \hat{v}_j^c, \hat{v}_j^{c'} \rangle_{c<c'}\Big] \quad (1)$$

$$\pi_j^{\text{odd}} = \Big[\det[\hat{v}_j^a, \hat{v}_j^b, \hat{v}_j^c]_{a<b<c}\Big] \quad (2)$$

$$\pi_j = \big[\pi_j^{\text{even}} \,\|\, \pi_j^{\text{odd}}\big] \quad (3)$$

with $\hat{v} = v/\|v\|$, pooled over joints by mean⊕max and concatenated with invariant bone lengths and a fixed set of anatomical signed volumes χ_τ (index-preserved, unpooled):

$$\Pi(h, \tilde{\boldsymbol{x}}) = \big[\,\mathrm{mean}_j \pi_j \,\|\, \max_j \pi_j \,\|\, (d_{ij})_{\mathcal{E}} \,\|\, (\chi_\tau)_{\mathcal{T}}\,\big]. \quad (4)$$

At the deployed widths (n_0=32, n_1=8) the parity-even subtotal is 160-d and the parity-odd completion adds 123-d (283-d total; full specification in the supplement). A liveness gate zeroes any generator touching a failed joint, which makes the dead-node *invariance* certified by E1–E8 exact under a known failure mask; the degradation reported in Tab. 4 instead uses the deployed gate-*off* model with no failure signal supplied, so the graceful fall-off measured there is message-passing dilution alone, not the gate.

Table 1. **The viewpoint guarantee is falsifiable.** Eight numerical gates on the deployed read-out: E1–E5 that the invariance is *present*, E6–E8 that it is the *intended* one (chirality live, laterality preserved). Roundoff-floor quantities are regime-stable, not bit-reproducible (representative fp64 run); E6–E7 are *modelling* quantities, not roundoff, reported for seed 0 of `certify_egru.json`. Parity/laterality gates report the even / chiral cut.

Gate	Criterion	Result
E1	rep. orthogonality $\|\rho^\top\rho - I\|_F$	$\leq 10^{-12}$ ✓
E2	motion-channel invariance	verified ✓
E3	rotation-sweep violation / slope	$\sim 10^{-16}$, $\lvert\cdot\rvert<0.5$ ✓
E4	fp32/fp64 violation ratio r	$> 10^8$ ✓
E5	translation invariance	$\sim 10^{-16}$ ✓
E6	parity (even / chiral)	0 / 4.4×10^{-2}
E7	laterality (even / chiral)	6.9 / 5.6×10^{-2}
E8	full-$SO(3)$, parity-odd active	$\sim 10^{-16}$ ✓

Proposition 1 (**$SO(3)$-invariance of the read-out**) *For an encoder that intertwines with the Wigner-D action on its latents, Π of Eq. (3)–(4) is invariant under $SO(3)$ for* arbitrary *weights; the predicted score is therefore invariant to any rigid motion of the camera. (Proof in the supplement.)*

Certificate. We audit the read-out with eight numerical gates (Tab. 1), all passed, each independently re-runnable and refutable (see Code availability): representation orthogonality $\leq 10^{-12}$; violation at the fp64 roundoff floor ($\sim 10^{-16}$) with a *flat* log–log slope in rotation angle ($|\text{slope}| < 0.5$, against ≈ 1 for a genuine modelling error) and an fp32/fp64 violation ratio $> 10^8$ (excluding a small modelling violation masquerading as noise); translation and read-out invariance both $\sim 10^{-16}$ (fp64) after the parity-odd channels are admitted. The guarantee is over Haar-random proper rotations of the *full* $SO(3)$, of which a camera azimuth or a cross-view displacement is one instance. A continuous-time (Neural CDE) variant, certified to the same standard, *fails* the mean-predictor floor and is reported only as a control (supplement).

4. Results

All experiments are pooled, subject-disjoint, over 3 seeds, except the TCN, ST-GCN and Ridge viewpoint curves of Fig. 2, which are 5-fold cross-validated at a single seed. *Protocol audit* (Tab. 2): we audit the point-cloud transformer baseline on the single-exercise slice most commonly reported. Honest epoch selection removes an 18.6% inflation (3 seeds) and leaves that model with no robust separation from a mean predictor — the paired subject-cluster bootstrap gives $\Delta = -0.44$ and clears zero in only 1/3 seeds. This is not subject leakage (each subject contributes one recording on that slice, so a random split is already

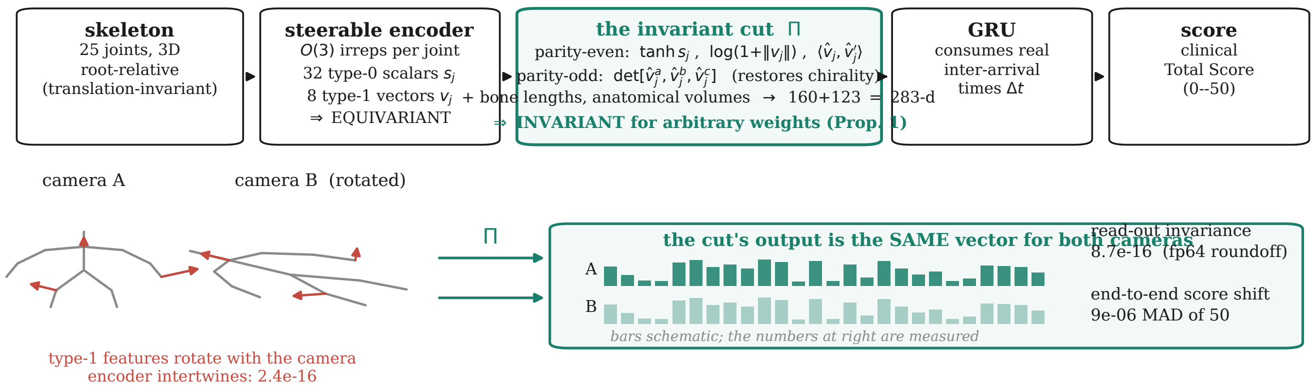

Figure 1. **The invariant cut is the method.** Top: the pipeline. Bottom: the same pose at two camera poses — the encoder's type-1 features rotate (equivariance), the cut's output does not (invariance, Prop. 1). Bar heights are schematic; the annotated quantities are measured (`certify_egru.json`, `final_tables.json`).

Table 2. Protocol audit of the *point-cloud transformer baseline*, single-exercise slice (KIMORE Ex. 1; subject-disjoint 5-fold; 3 seeds). *Honest* selects the epoch on a held-out validation fold; *inflated* takes the per-epoch test minimum (the reference protocol). Δ is the paired subject-cluster bootstrap MAD(honest) − MAD(floor); an interval below 0 means the model beats the mean predictor. The signal here is *seed-fragile*, clearing zero in only 1/3 seeds; pooling fixes this, for this model and every other (Tab. 3).

Seed	Honest / Inflated	Selection bias	Δ vs. floor (95% CI)
0	6.90/5.45	+1.44 (20.9%)	+0.20 [−0.59, +1.03]
1	6.53/5.00	+1.54 (23.5%)	−0.46 [−1.60, +0.72]
2	5.84/5.17	+0.67 (11.4%)	**−1.07** [**−1.98**, **−0.15**]
mean	6.42±0.44/5.21±0.19	+1.22 (18.6%)	−0.44

Table 3. Clean-data accuracy (pooled, subject-disjoint, mean±std over 3 seeds). The 0.33 MAD floor is the *fixed-configuration* one; the wider seed-to-seed spread (0.48) is in the supplement. Δ vs. floor is the paired subject-cluster bootstrap of Tab. 2 (n=77 subjects, checkpoints re-evaluated without retraining; `pooled_bootstrap_eval.py`). Spearman ρ is computed per seed on that seed's complete out-of-fold vector, pooled over exercises and again within each exercise.

Model	Group	MAD ↓	Params	Spearman ρ ↑ pooled	Spearman ρ ↑ within-ex.	Δ vs. floor (95% CI)
PCT (baseline)	—	**6.46 ± 0.20**	4.91M	0.60 ± 0.02	0.55 ± 0.02	−1.87 [−2.72, −1.06]
PCT + rotation aug.	—	7.47 ± 0.23	4.91M	—	—	—
InvariantGRU (hand-crafted)	$SO(3)$	**6.31 ± 0.16**	0.21M	0.59 ± 0.01	**0.56 ± 0.01**	−2.03 [−2.89, −1.23]
EGRU (ours)	$SO(3)$	**6.73 ± 0.30**	0.66M	0.56 ± 0.04	0.52 ± 0.04	−1.60 [−2.37, −0.88]
Mean-predictor floor	—	8.31 ± 0.05	—	0.17 ± 0.01	−0.07 ± 0.02	—

subject-disjoint); the slice simply carries no robust subject-level signal, which is why we pool the five exercises and gate every experiment on a per-exercise mean-predictor floor. Pooled, *every* model here clears that floor (Tab. 3).

The clean-accuracy metric is saturated. EGRU (6.73 ± 0.30), a GRU on hand-crafted invariants (InvariantGRU, 6.31 ± 0.16), and the point-cloud transformer (PCT, 6.46 ± 0.20) are mutually indistinguishable: every gap is inside the 0.33 MAD seed floor (Tab. 3). This is a statement about the models *relative to each other*, not about whether any of them carries real subject-level signal: unlike the fragile single-exercise slice (Tab. 2, clears zero in only 1/3 seeds), the pooled bootstrap Δ vs. floor is entirely below 0 for all three — PCT $-1.87\,[-2.72, -1.06]$, InvariantGRU $-2.03\,[-2.88, -1.23]$, EGRU $-1.60\,[-2.37, -0.88]$ — so the tie is a tie among models that each genuinely beat the mean predictor, not three models indistinguishable from noise. The tie survives the metric change: Spearman ρ separates the same models by at most 0.05 (Tab. 3), so saturation is a property of the benchmark, not of the error measure chosen to report it. ρ also needs its null stated — the mean predictor scores pooled $\rho = 0.17$ while ignoring its input entirely, and exactly 0 once the between-exercise score differences it exploits are removed. A benchmark on which three architectures of vastly different inductive bias cannot be told apart is not measuring architecture — which is the premise for evaluating on an axis that can. Restoring chirality ($O(3) \to SO(3)$) costs +0.11 MAD for EGRU — inside the floor — for 16.7% more parameters and the removal of a silent handedness assumption; we adopt the chiral model on that basis, not an accuracy one. The chiral read-out's wider seed spread (±0.30 vs ±0.08 for the parity-even model) reflects the added parity-odd capacity, not instability — every seed stays inside the floor.

One asymmetry against the baseline is worth closing: EGRU sees a one-hot exercise id and the floor is per-exercise, so pooled PCT alone was scored blind to the exercise. Re-training it with identical conditioning (3 seeds, both arms) changes nothing: clean 6.59 ± 0.07 vs. 6.46 ± 0.20, augmented 7.42 ± 0.22 vs. 7.47 ± 0.23 — inside the floor and opposite in sign — and the per-sequence swing is if anything *larger* (21.43 vs. 20.12 MAD worst sequence). Viewpoint fragility is not an artefact of withholding exercise identity; we report the unconditioned arm throughout.

Viewpoint invariance is a theorem, not a fit (Fig. 2). Across test-time camera azimuth the certified-invariant models are flat to machine precision (EGRU mean 9×10^{-6} MAD, worst single sequence 3×10^{-4}; InvariantGRU exactly 0; hand-crafted invariant features Ridge at the fp64 roundoff floor, 10^{-14} — both invariant by construction, so these two curves are a positive control, not independent evidence), while all non-equivariant baselines degrade catastrophically: PCT crosses the mean-predictor floor between 30° and 45° (mean per-sequence shift 9.33 MAD at its worst angle, worst sequence 33.96 of fifty); TCN 13.17 MAD; ST-GCN 9.18 MAD. The pattern holds across three independently-designed non-equivariant architectures (temporal-conv, graph-conv, attention): viewpoint fragility is generic to non-equivariance, not PCT-specific. Rotation augmentation flattens PCT's aggregate curve but leaves a *per-sequence* score shift averaging 3.01 MAD — and the average is the wrong statistic for a clinical claim: the 95th percentile is 8.65 MAD and the worst single held-out sequence moves 20.12 (a global maximum over all 380 held-out sequences × 3 seeds; the fold-averaged max would read a flattering 12.81), two fifths of the scale and 2.01 between-subject SD, purely because the camera moved. This is the difference between a guarantee and a fit. It is not an undertraining artefact: at 3× and 9× the epoch budget the shift stays flat (3.01/3.27/2.94 MAD mean at 60/180/540 epochs, worst 20.1/21.4/19.9), so it is what augmentation converges to — though we varied the budget, not augmentation strength, and a differently tuned schedule may narrow the gap.

On NTU the guarantee is measured, not inherited. A Haar-random $SO(3)$ rotation of all 18,674 Cross-View test clips changes *not one* prediction (agreement 100.00%, Top-1 82.95% unchanged; drift 2.3×10^{-13} in fp64, 5.7×10^{9} below fp32 — roundoff), where ST-GCN agrees on 91.89%, loses 3.03 points, with a drift precision cannot move (ratio 1.0008): structural, not numerical. Both are within-model. But a rotated skeleton is not a relocated camera: NTU films each performance at three cameras *at once*, and on those 18,674 triples — three skeleton *estimates* of one event — our agreement falls to 73.03% against ST-GCN's 73.86%: a small but resolvable deficit (paired exact binomial, 5,394 discordant triples, p=0.037), still confounded by the protocol caveat below. The theorem removes a viewpoint change's geometry exactly and none of the tracker noise a relocation adds; reporting the rotation alone would overstate it. NTU otherwise contributes generalisation: trained *without* rotation augmentation on cameras 2–3 and tested on camera 1, the encoder reaches 82.98% Top-1 (T=100, single-clip, one seed) — also why it sits below the $\sim 88\%$ for this split. We give no ST-GCN accuracy control: our runs differ in sampling, view-normalisation and optimiser, so the pair would not compare.

Clinical interpretation of the MAD scale. No inter-/intra-rater reliability coefficient (ICC, κ) is reported for

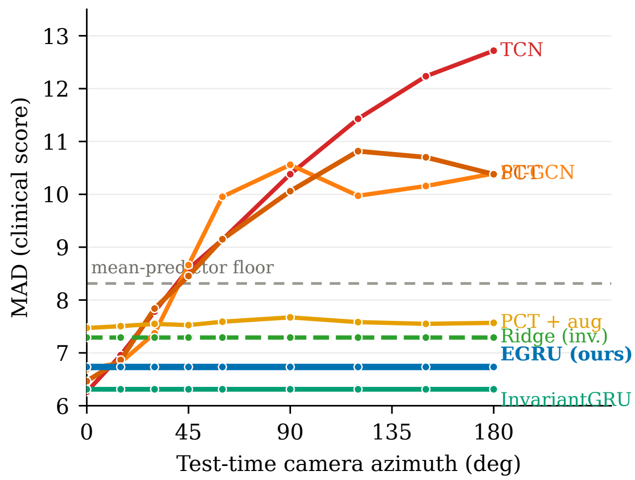

Figure 2. MAD vs. camera azimuth. Certified-invariant models (EGRU, InvariantGRU, hand-crafted Ridge) are flat by theorem; all non-equivariant baselines (PCT, TCN, ST-GCN) degrade 9–13 MAD at extreme angles — viewpoint fragility is generic to non-equivariance. Augmentation flattens PCT's aggregate but not its per-sequence swing (mean 3.01 MAD, worst sequence 20.12). Azimuth here is a rigid rotation of the ground-truth Kinect skeletons; it isolates the geometric effect of viewpoint and does not model per-view tracker degradation, which we measure separately on a physically relocated camera (supplement).

KIMORE's clinical Total Score [1] or for REHAB24-6's; a targeted search, including direct attempts to retrieve the Capecci et al. full text (both the publisher record and the indexed copy returned access-blocked to automated retrieval), could not surface one, and we do not fabricate one — we flag its absence as this paper's clearest open measurement gap. As external context, *movement-quality rating* — the construct KIMORE's clinicians perform, not a functional-recovery outcome — shows moderate-to-good inter-rater agreement (Cutting Movement Assessment Score ICC 0.58–0.91; cross-diagnostic Movement Quality Score 0.93): a defensible band on what a KIMORE-specific coefficient would show, not a substitute for one. As a magnitude-of-change cross-check from a different construct and scale, minimal clinically important differences (MCIDs) on comparable 0–66/0–100-point motor-*recovery* scales sit in the low single digits to low teens (Fugl-Meyer upper-extremity 4–12.4, lower-extremity 6, Stroke Rehabilitation Assessment of Movement 1.9–4.8 per subscale), putting the augmented baseline's per-sequence swing (mean 3.01, worst sequence 20.12, of 50) in the same order of magnitude as an established change-significance threshold on cognate instruments — at the tail, above it — and our own score shifts under rotation (9×10^{-6}), int8 and streaming (0.05 and lower, Sec. 5) orders of magnitude below it. A camera relocation that moves a patient's score by more than an MCID is a measurement the instrument cannot be trusted to make; that is the practical content of the guaran-

Table 4. Sensor-node failure: MAD as k nodes freeze (mean over 3 seeds × 5 folds; identical hash-locked inputs; no retraining). Underline: past the floor (8.31). "+oracle": InvariantGRU given an oracle liveness mask (same checkpoint; features touching the dead joint zeroed at inference) — the measured fairness control.

Dead nodes k	EGRU (ours)	InvariantGRU	+oracle	PCT
1	7.10	$\underline{9.78}$	$\underline{9.39}$	6.76
4	$\underline{8.41}$	$\underline{15.96}$	$\underline{15.25}$	7.51
8	$\underline{10.50}$	$\underline{18.37}$	$\underline{16.33}$	$\underline{8.73}$
MAD lost	+3.76	+12.05	+10.01	**+2.27**

tee. The sharper, in-dataset question — does a 3.01-point swing cross KIMORE's clinical bands? — has no answer: those bands do not exist, the five groups being recruitment categories, with a control scoring below a patient on every exercise (Es5: Expert 25.0, Parkinson 50.0). Measured against the score's actual resolution, the mean swing is 0.30 between-subject SD and the worst sequence 2.01 SD, and it exceeds the gap separating 20.2% of subject pairs, 83.0% at the tail — pairs a camera can reorder (supplement). The MCID band then reads as corroboration from adjacent instruments, not as a KIMORE-specific threshold and not as the load-bearing claim.

Does the guarantee cost robustness? Sensor-node failure (Tab. 4). This is a tax check, not a selling point. Freezing k tracking nodes at their first-frame pose, the hand-crafted arm collapses through the floor at a *single* dead joint (6.31 → 9.78) because its features name their joints; EGRU pools learned graph messages and loses +3.76 MAD over 8 dead joints. **PCT wins this axis outright** (+2.27): it leads on KIMORE and, on REHAB24-6, overtakes us between k=2 and k=4 and finishes 0.021 AUROC ahead at k=8 (supplement). Node robustness is not a property of equivariance and we do not claim it. The table supports something narrower: the guarantee is not paid for with a robustness collapse (k=8 crosses the floor for every model, EGRU included). Both models see identically corrupted joints with no imputation, and we measured the fairness objection rather than assuming it: an *oracle* liveness mask (told which joint died, no retraining) closes only ∼25% of the gap to EGRU (+12.05 → +10.01 MAD; Tab. 4, right) — the graph encoder needs no such signal and loses far less, so the gap is architectural, not a missing-information artifact. The mechanism is invariant to the failure model (five operators incl. lag/burst/axis/lateral; supplement).

The mechanism is not an artifact of the freeze model. Under four further operators — stuck-at-lag, bursts, single-axis depth noise, left-limb-only — the ordering is invariant: the hand-crafted arm degrades most and the graph-pooled encoder least among the two exact-invariance models (Tab. 5). *Persistent* faults (freeze, lag) hurt naming features far more (+12.05, +8.88) than *transient* ones (burst +2.83, axis +1.93) — a length-masked temporal mean dilutes an intermittent spike but not a permanently corrupted coordinate — and unilateral failure does not break the parity channels (InvariantGRU +8.73, EGRU +3.43).

Table 5. Node failure under five operators (chiral models, 3 seeds × 5 folds, hash-locked). Each cell: MAD at k=1 / k=8 and total lost. The freeze/random row reproduces Tab. 4.

Failure operator	EGRU (ours)	InvariantGRU	PCT
freeze (random)	7.10/10.50 (+3.76)	9.78/18.37 (+12.05)	6.76/8.73 (+2.27)
freeze (left-only)	7.11/10.16 (+3.43)	10.81/15.04 (+8.73)	6.83/9.62 (+3.15)
stuck-at-lag	6.82/7.76 (+1.03)	9.02/15.19 (+8.88)	6.60/7.11 (+0.64)
sporadic burst	6.74/6.78 (+0.05)	6.69/9.14 (+2.83)	6.47/6.45 (−0.02)
axis/depth noise	6.77/7.20 (+0.47)	6.70/8.24 (+1.93)	6.48/6.50 (+0.03)

Table 6. Does each model clear the floor (8.31 MAD) under each stress? The node column is k=2; the full sweep is Tab. 4, where ours first crosses the floor at k=4 and the point-cloud baseline at k=8. Last column: *mean* per-sequence score shift at the worst camera azimuth; the tail it conceals is in the text. EGNN shares the identical cut (viewpoint exact by construction). †Canon-PCA's near-zero is *empirical*, a frame estimate stable on KIMORE, not a certificate.

Model	Params	Clean	90°	2 dead	Mean degr.
EGRU (ours)	0.66M	6.73	6.73✓	7.55✓	$\mathbf{9\times10^{-6}}$
InvariantGRU	0.21M	6.31	6.31✓	11.64×	**0**
Lighter $E(n)$ (EGNN)	∼0.6M	6.88	6.88✓	8.44×	1.4×10^{-5}
Canon-PCA + PCT	4.91M	6.85	6.85✓	7.79✓	≈0†
PCT (baseline)	4.91M	6.46	10.06×	7.10✓	9.33
PCT + rot	4.91M	7.47	7.67✓	7.76✓	3.01

The whole thesis in one grid (Tab. 6). *No model wins all five columns*, and that — not any single stress — is the claim. PCT is both the most accurate clean model (6.46) and the most node-robust (7.10); InvariantGRU is the cheapest at exactly zero degradation yet collapses at two dead nodes (11.64); and three models clear every stress — ours, Canon-PCA+PCT, and rotation-augmented PCT — so we are not the only survivor. Among those three, ours alone runs at 0.66M rather than 4.91M parameters and alone rests on a theorem rather than an estimated frame or an augmentation tolerance. What separates the survivors is the last column (Fig. 3a): all flat in aggregate, but the augmented baseline's mean per-sequence degradation is 3.01 MAD (worst sequence 20.12) against our 9×10^{-6} (3.0×10^{-4} worst), at 7.4× more parameters. This is the only categorical difference rather than a trade: 3.01 is a tolerance that happens to be small on this data and has a 20 MAD tail; 9×10^{-6} is roundoff around a value the theorem fixes at zero. Every other column is an axis we may lose, and on node failure we do. Behind the *identical* cut, a lighter $E(n)$ (EGNN) encoder ties clean accuracy and viewpoint (exact by construction) but is more node-fail brittle: tuned across depth, width, and coordinate-clamp over three seeds it loses $+6.4 \pm 1.3$ MAD over eight dead joints (vs. +3.76) and crosses the floor already at two (8.44 vs. 7.55) — the steerable encoder is the more robust equivariant option.

Canonicalization ties us offline — and the difference

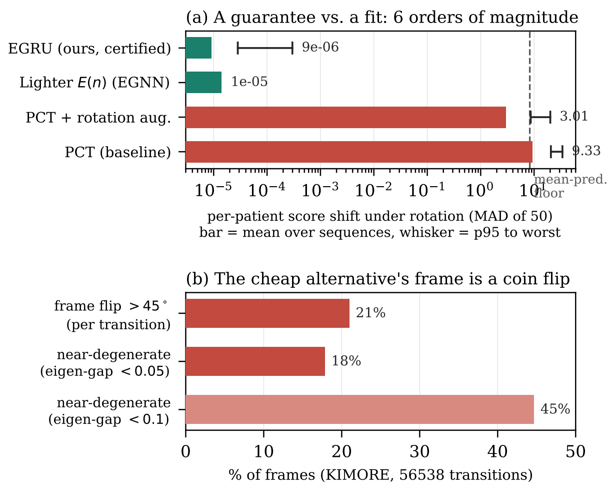

Figure 3. **The axis that separates the survivors.** (a) Per-patient score shift under camera rotation (log scale). Bar is the mean over held-out sequences at the worst azimuth; whisker runs 95th percentile to worst sequence. EGNN has no measured tail and carries no whisker. (b) PCA frame instability: frame transitions flipping $> 45°$ (21%), and frames within an eigen-gap of 0.05 of degeneracy (18%).

is the guarantee. The strongest objection to an equivariant model is that a non-equivariant backbone fed a canonical frame reaches the same place for less. It does, here: a per-frame PCA principal-axis frame feeding the transformer (Canon-PCA+PCT, Tab. 6) matches every offline axis — clean 6.85 MAD, 7.79 at two dead joints, and a viewpoint invariance empirically exact for arbitrary $SO(3)$, not merely the azimuth a clinician sweeps (worst canonical-coordinate shift 3.7×10^{-11} over 32 random rotations). This is the baseline's strong form: axes sign-disambiguated by a deterministic rule (each oriented so its dominant joint projection is positive) and the frame forced right-handed, so the discrete $\pm$ ambiguity is resolved before any number below is measured.

The difference is not accuracy but *conditioning*. Under a clean rotation the PCA frame is exactly equivariant — rotating the input sends the covariance to $RCR^\top$, whose eigenvectors are exactly RV — and we measure that residual at 3×10^{-6} degrees. But by Davis–Kahan [16] an estimated eigenvector subspace moves by $O(\|E\|/\gamma)$ under a perturbation E with eigen-gap γ, and a depth sensor supplies $E \neq 0$ at every frame. On KIMORE γ is small far more often than not — 18% of frames within 0.05 of degeneracy, 45% within 0.10, a standing body being nearly axially symmetric — so amplification is the common case, not a corner. Injecting camera-frame noise at 2% of body scale and comparing the frame recovered at two camera poses, the residual rotation grows from $1.2°$ where $\gamma > 0.25$ to $11.1°$ where $\gamma < 0.02$: a $9\times$ amplification, with 14.5% of frames past $45°$.

A better sign rule does not fix it. Replacing the dominant-joint convention with a third-moment rule leaves the amplification intact ($12.1°$, trimming only catastrophic flips to 11.1%), because a sign rule resolves a *discrete* ambiguity whereas degeneracy is a *continuous* one. Temporal smoothing is worse ($28.8°$; 31.6% past $45°$): a frame locked onto the wrong branch persists, and the frame becomes history-dependent, trading per-frame equivariance for lag. A learned point-estimate frame inherits the same obstruction — no canonicalizer is simultaneously exactly equivariant and continuous where the configuration's stabilizer is non-trivial [17]. We report this at the frame rather than the score: the frame measurement is model-free and binds any spectral canonicalizer, whereas how much wobble survives to the output is corpus-dependent and secondary to whether the mechanism exists. Our cut estimates *no* frame: it is $SE(3)$-invariant for arbitrary weights by construction (Proposition 1), certified to roundoff over the full rotation group — so the *same* certificate covers, without retraining, NTU RGB+D Cross-View's displaced $45°$ camera. A frame estimator offers neither the bound nor that transfer. We claim no accuracy win here — the tie is total; we claim a *certified* worst case they structurally cannot offer, and recommend the cheaper models where an off-distribution guarantee is not required.

When the worst case is the deployment case. A home device is set up by the patient and left unattended: the camera sits wherever a phone stand or laptop lid lands, and the consumer depth sensor is the component most likely to lose a joint. Neither cheaper alternative is safe against *both* conditions at once. Canon-PCA+PCT's viewpoint guarantee is only empirical — Fig. 3b puts the PCA frame within an eigen-gap of 0.05 of degeneracy on 18% of frames, where a flipped frame is silent: no flag, just a wrong score. InvariantGRU's *is* exact like ours, but its named-joint features make it the most node-fail-fragile model measured: one frozen node drives it from 6.31 to 9.78 MAD, through the floor (Tab. 4), the freeze mode a consumer sensor exhibits routinely. Those are exactly the two conditions a home device cannot rule out in advance, and EGRU is the only model measured provably safe against both — not that the others are worse in a fixed-rig clinic, but that neither offers a bound surviving unsupervised deployment.

Second-corpus replication (REHAB24-6). We replicate the structural story on REHAB24-6 (binary correct/incorrect repetitions; 1057 reps, 10 subjects; the identical 0.66M model as a BCE logit), an AUROC task over 3 seeds and subject-disjoint CV (Tab. 7). The accuracy tie holds — EGRU 0.738 ± 0.01 versus the transformer 0.707 ± 0.02 — and, decisively, the viewpoint theorem transfers: under a $90°$ camera rotation EGRU's AUROC is *unchanged* (logit drift $\leq 1.5 \times 10^{-4}$), while the transformer collapses to chance (0.52). Node failure degrades

Table 7. REHAB24-6 second-corpus replication (3 seeds, subject-disjoint CV). AUROC@90° is the same model evaluated under a 90° camera rotation: EGRU is invariant by theorem (logit drift $\leq 1.5 \times 10^{-4}$); the transformer falls to chance (drift 23.2). Node-fail: AUROC lost over 8 frozen nodes. The two models *cross* between 2 and 4 frozen joints; full curve and fold-level spread in the supplement.

Model	AUROC	AUROC@90°	Node-fail lost
EGRU (ours)	$\mathbf{0.738 \pm 0.01}$	$\mathbf{0.738}$	$+0.14$
PCT (baseline)	0.707 ± 0.02	$0.52\times$	$\mathbf{+0.08}$

both gracefully. The guarantee is thus not a KIMORE artifact — though at 10 subjects this corpus replicates *structural* properties, not accuracy: a certified invariance either transfers exactly or it does not, and we draw no accuracy conclusion from it.

Per-family contribution of the cut. Zeroing one family of Eq. (3) at inference on the deployed chiral model (dimension preserved, no retraining) resolves the mechanism to family granularity (full table in the supplement). Two families carry accuracy — the pairwise *cosines* (+1.83 MAD to remove) and the learned *triple products* (+2.75) — and removing the cosines is doubly harmful, inflating node-failure degradation the most (+7.29). The parity-odd families *split*: the learned triple products are relied upon while the fixed anatomical volumes are inert (+0.19), which reconciles rather than contradicts the chirality null — that null is a *retrain* comparison (parity-even families substitute during training), whereas this is an *inference* ablation on a model trained *with* the family.

Two negative results, honestly — each a mapped boundary of where structure pays. (i) Native consumption of irregular arrivals does not help on KIMORE — the fixed-grid baseline is better at every drop level under a hash-locked budget — and a spectral census [23] predicts it: 97.7% of *positional* energy lies below the resampling corner (though 70% of *velocity* energy lies above), so resampling preserves pose geometry while low-passing the derivative band — on KIMORE's healthy-form exercises the discriminative signal is positional, so this acts as a denoiser; the boundary would move for tremor-dominated tasks. The mechanism is real (a 5 Hz tremor is recovered at r=1.000 through 70% drops where a resampled estimator decays to 0.785), so the claim is a scope boundary, not a defeat. (ii) Restoring chirality is principled but not rewarded on healthy-form exercises with no chiral pathology — the boundary at which a handedness prior stops paying. Both are stated with the corpus properties that would reward them (supplement).

5. Deployment Analysis

We treat the guarantee as a systems property (Tab. 8). *Precision budget:* the invariance floor climbs monotonically

Table 8. Deployment audit. Top: the *full* equivariance precision budget of the deployed pooled EGRU (INV floor = relative violation of the 160-d parity-even projection; score shift in MAD of 50). fp16 is 3.2× tighter than bf16 — equivariance pays in mantissa bits, which bf16 trades for exponent range — and the int8 cliff is *solely* activation grid-rounding (weight-only quant preserves the theorem). Bottom: causal streaming vs. bidirectional read-out.

Arithmetic	INV floor	Score shift	Note
fp64	5.91×10^{-16}	2.2×10^{-14}	exact (theorem)
fp32	2.68×10^{-7}	8.9×10^{-6}	= main viewpoint
fp16	2.62×10^{-3}	0.024	> bf16
bf16	8.34×10^{-3}	0.195	mantissa cost
int8, weights only	3.09×10^{-7}	1.2×10^{-5}	theorem holds
int8, activations only	3.38×10^{-2}	0.056	cliff (acts)
int8, weights+acts	$\mathbf{3.40 \times 10^{-2}}$	**0.051**	weight-quant. theorem-safe
Streaming: 8.9 ms/frame (112 fps); read-out cut unchanged; causal vs. bidir NTU X-View 79.8 vs. 83.0 (−3.2 pt).			

from fp64 to int8, and the int8 cliff is attributable *solely* to activation grid-rounding of the steerable channels — weight quantization preserves the theorem exactly (a quantized model is a different equivariant model) — while the reported clinical score still moves only 0.05 MAD — below the floor, 0.005 of the between-subject SD, and one to two orders of magnitude below the comparator MCID band established in Sec. 4. *Streaming:* because invariance lives in a per-frame encoder and an invariant cut, the read-out converts to a strictly causal unidirectional form with a running-mean pool that emits a first score 8.9 ms after the first frame on an RTX-class GPU (batch 1; 112 fps, inside a 30 fps budget) — we report GPU latency as an upper bound on compute and leave edge-SoC profiling to deployment — reaching a usable prediction after $\sim$55 frames (median), an honest time-to-first-*reliable*-score for real-time use. The invariant read-out cut is unchanged by the causal pooling, so Proposition 1 carries over verbatim; the cost is 3.2 points of NTU accuracy vs. the bidirectional model (NTU-60 X-View SOTA is $\sim$92%; our backbone is deliberately untuned — we measure the streaming *delta*, not absolute accuracy). A canonicalization front-end would instead add a per-frame eigendecomposition to the hot path and re-open the invariance question after quantization; here the guarantee is a fixed property of the read-out and survives both int8 and causal streaming without re-certification. *A real camera move:* replaying one clip to a webcam from seven physical camera poses, our score moves 0.74 of fifty on average (worst 1.41) against PCT's 2.16 (worst 3.25), a gap above 2× under lag and frame-rate perturbation. One subject, one exercise, and it bounds *shift*, not accuracy — webcam skeletons are off-distribution for both — and, as on NTU's three simultaneous cameras, ours is not zero (supplement).

6. Conclusion

On the metric the field reports, no architecture separates from another: on clean frontal KIMORE our model, the baseline, and a hand-crafted-invariant GRU tie within a 0.33 MAD nondeterminism floor — the metric is saturated. What separates them is off the frontal manifold. Our readout is $SE(3)$-invariant by construction and certified to machine roundoff, so relocating the camera moves a patient's score by 9×10^{-6} MAD for arbitrary weights, while a broken tracking node — fatal to the hand-crafted model — is absorbed by the graph encoder; both properties replicate on a second corpus, and the guarantee survives to int8 and to causal streaming. We recommend the field report a mean-predictor floor and the per-sequence degradation *distribution* — not only its mean, which on our own augmented baseline understates the worst patient by nearly 7× — alongside aggregate error, and evaluate off the frontal manifold: camera relocation and node failure are ordinary conditions of a home device, not adversarial constructions.

Code availability. KIMORE [1], NTU RGB+D and REHAB24-6 are public. Our certificate, harnesses, benchmarks, and the script regenerating every number here from banked artifacts will be released on acceptance; the URL is withheld for double-blind review.

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